English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Computer Science, 8 June 2024 Test 2 ——— *The exact grading formula depends on the results. As an approximation, use 2 + the number of points. Time allowed: 3 hours. Good luck!*

Problem 1. Based on previous games, Angel models his score with mean 5011 points and variance 4000. Approximately how many games are needed for him to be at least 99% certain that:

1. (0.5 points) the total score is at least one million;
2. (0.5 points) his highest score among those games is at least 10000?

Problem 2. (1 point) Let ξ, η be independent, $\xi \sim \text{Exp}(2)$ and $\eta \sim U(0,3)$, with densities

$$f_\xi(x) = \begin{cases} 2e^{-2x}, & x > 0, \\ 0, & \text{otherwise}, \end{cases} \qquad f_\eta(x) = \begin{cases} 1/3, & 0 < x < 3, \\ 0, & \text{otherwise}. \end{cases}$$

Find their correlation, $\mathbb{P}(\xi < \eta)$ and the density of ξ/η.

Problem 3. Let X, Y be independent exponential variables with rates $1/\alpha$, $1/\beta$, respectively, and let $X_1, \ldots, X_n$ be independent observations of X.

1. (0.25 points) Let $\overline{X} = (X_1 + \cdots + X_n)/n$. For large n, find c with $\mathbb{P}(|\alpha - \overline{X}| > c) \leq 1\%$.

Let $Z = \sqrt{X/Y}$ and let $Z_1, \ldots, Z_n$ be independent observations of Z.

2. (0.75 points) Is it true that
$$\frac{Z_1 + \cdots + Z_n}{n} \xrightarrow[n\to\infty]{\text{a.s.}} \sqrt{\frac{\mathbb{E}[X]}{\mathbb{E}[Y]}}?$$
If so, prove it; otherwise find the limit.

Problem 4. The aim is to find integer moments of $X \sim N(\mu, \sigma^2)$. Begin with $Z \sim N(0,1)$.

1. (0.25 points) Find $\mathbb{E}[e^{\lambda Z}]$ for $\lambda > 0$ and $\mathbb{E}[Z^k]$ for $k \leq 6$.
2. (0.25 points) Prove $\mathbb{E}[f(Z)Z] = \mathbb{E}[f'(Z)]$ when both expectations are well defined.
3. (0.25 points) Find $\mathbb{E}[Z^k]$ for every $k \geq 0$. Can you obtain the corresponding result for $\mathbb{E}[X^k]$?
4. (0.25 points) Find $\mathbb{P}(Z^k > Z^{k-1})$ for $k \geq 5$.